DNA Methylation Analysis of Cerebral Small Vessel Disease Using Illumina EPIC Data

1. Abstract

This study investigates genome-wide DNA methylation differences between cerebral small vessel disease (SVD) patients and healthy controls using Illumina Infinium MethylationEPIC array data (GSE318387). Raw IDAT files were processed using the **minfi** package in R, followed by normalization using quantile normalization. Differential methylation analysis was performed using the **limma** framework with **Benjamini–Hochberg false discovery rate (FDR)** correction. A total of **7,295 significantly differentially methylated CpG sites (adjusted p-value < 0.05)** were identified. Visualization using PCA, heatmap, volcano plot, and beta-value distribution confirmed clear epigenetic differences between groups. The findings suggest that DNA methylation alterations may contribute to SVD pathogenesis and could serve as potential biomarkers.

2. Introduction

DNA methylation is a key epigenetic mechanism regulating gene expression without altering the underlying DNA sequence. It primarily occurs at CpG dinucleotides and plays a crucial role in cellular differentiation, genomic stability, and disease development.

Cerebral small vessel disease (SVD) is a major cause of stroke and cognitive decline. While genetic factors contribute to disease susceptibility, **epigenetic modifications such as DNA methylation provide an additional regulatory layer** that may explain disease heterogeneity.

This study aims to:

- Identify **Differentially Methylated Positions (DMPs)** between SVD and control groups
- Understand epigenetic alterations associated with disease
- Provide insights into potential biomarkers and regulatory mechanisms

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3. Materials and Methods

3.1 Dataset

- **Dataset:** GSE318387
- **Platform:** Illumina Infinium MethylationEPIC (850K array)
- **Samples:** 32 (16 SVD, 16 Controls)
- **Source:** Peripheral blood

3.2 Tools and Software

- R Statistical Software
- minfi (data preprocessing)
- limma (differential analysis)
- pheatmap (visualization)

3.3 Methodology

1. Raw IDAT files were downloaded and imported using *minfi*
2. Quality control was performed to ensure data integrity
3. Normalization was applied using **quantile normalization (preprocessQuantile)**
4. Beta values (0–1) were computed to represent methylation levels
5. Sample groups (SVD vs Control) were defined using metadata
6. Differential methylation analysis was performed using **limma linear modeling**
7. P-values were adjusted using **Benjamini-Hochberg FDR correction**
8. Significant CpGs were selected (adjusted p-value < 0.05)
9. Visualization:
    - Beta value distribution (boxplot)
    - PCA (sample clustering)
    - Heatmap (top CpGs)
    - Volcano plot (significance vs effect size)

4. Results

4.1 Data Overview

- Total CpG sites analyzed: ~930,000
- Samples: 32
- Groups: Control vs SVD

4.2 Differential Methylation Analysis

- **Significant CpGs:** 7,295

- **Criteria:** Adjusted p-value < 0.05 (FDR corrected)

These CpG sites represent loci with statistically significant methylation differences between SVD and control groups.

4.3 Hypermethylation vs Hypomethylation

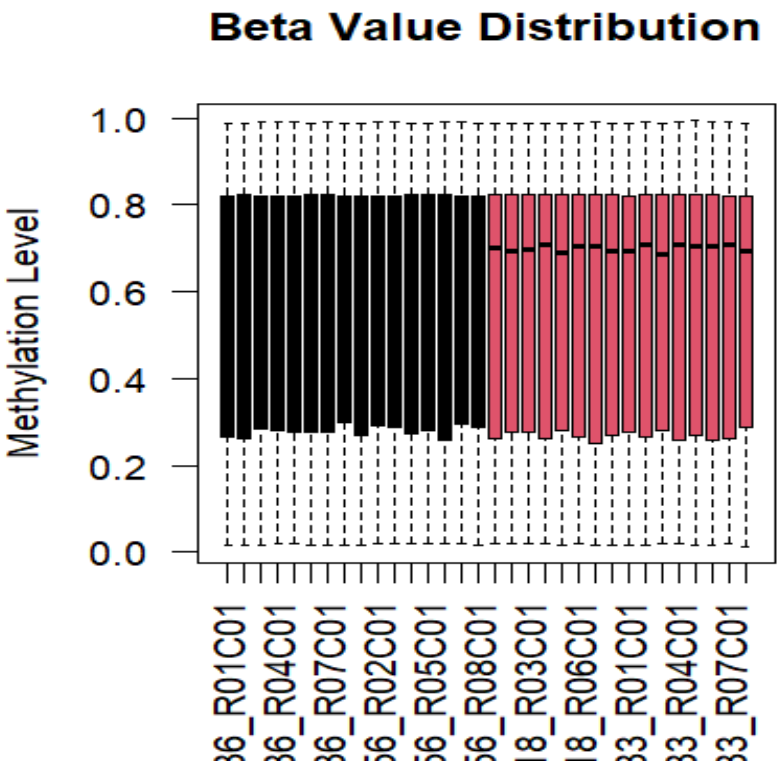
- **Hypermethylated CpGs (logFC > 0):**
  Increased methylation in SVD → may lead to **gene silencing**
- **Hypomethylated CpGs (logFC < 0):**
  Decreased methylation in SVD → may lead to **gene activation**

This indicates **both gene repression and activation mechanisms** are involved in SVD.

4.4 Visualization and Interpretation

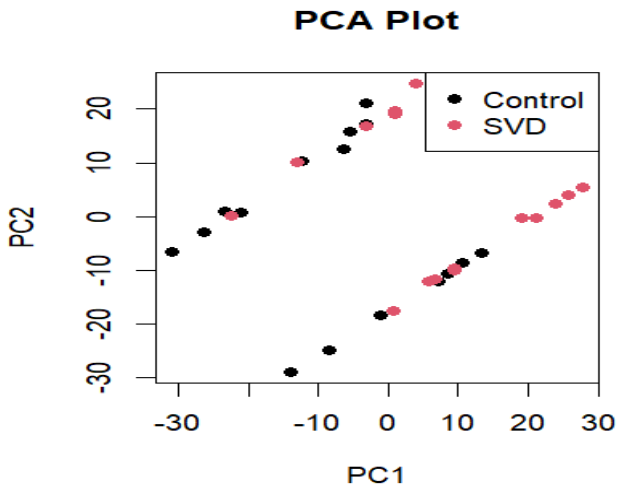
- **Beta Value Distribution (Boxplot)**

  - Shows uniform distribution across samples
  - Confirms **successful normalization**
  - No major batch effects observed



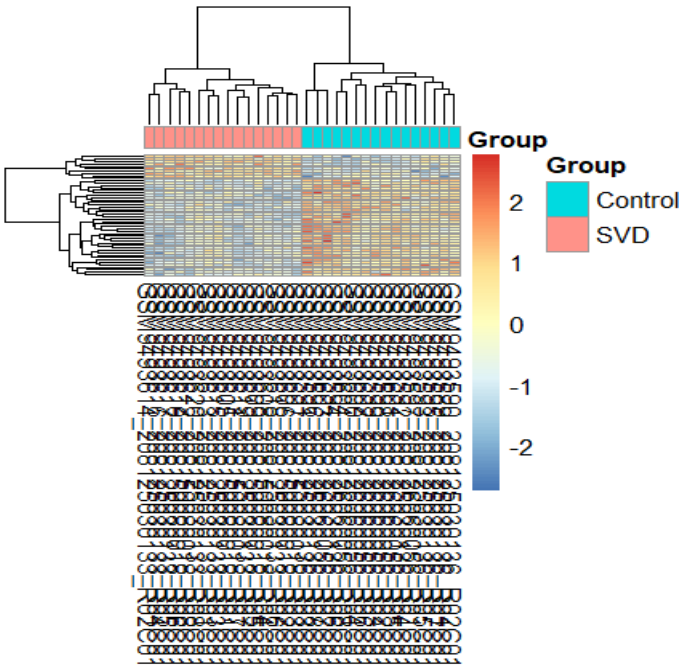
🔹 PCA Plot

- Partial separation between SVD and Control groups
- Indicates **systematic epigenetic differences**
- Some overlap suggests biological variability



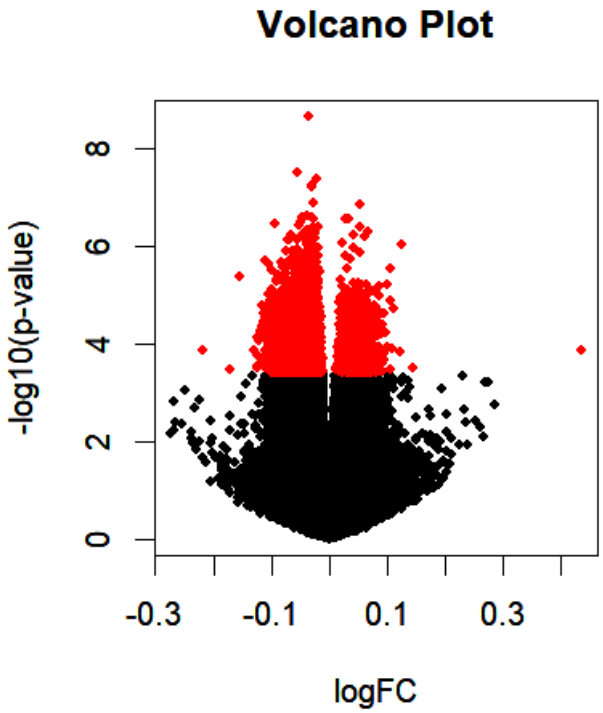
🔹 Heatmap

- Clear clustering of samples by group
- Demonstrates distinct methylation signatures
- Supports reliability of differential analysis



- **Volcano Plot**

  - Highlights significantly altered CpGs
  - Red points indicate statistically significant changes
  - Symmetry suggests both hyper- and hypomethylation



5. Discussion

This study identified **7,295 differentially methylated CpG sites**, indicating substantial epigenetic differences between SVD and control groups.

The presence of both hypermethylation and hypomethylation suggests:

- **Gene silencing (hypermethylation)** may suppress protective genes
- **Gene activation (hypomethylation)** may activate pathological pathways

The clustering patterns observed in PCA and heatmap confirm that methylation profiles can distinguish between disease and healthy states.

These findings align with the growing evidence that **epigenetic dysregulation contributes to neurological diseases**, including vascular-related brain disorders.

6. Limitations

- **Small sample size (n=32)** may limit statistical power
- Blood samples may not fully reflect **brain-specific methylation patterns**
- Only **DMP analysis** was performed (no DMR analysis)
- Lack of gene/pathway enrichment analysis

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7. Future Work

- Perform **DMR (Differentially Methylated Region) analysis**
- Conduct **gene annotation and pathway enrichment analysis**
- Validate findings using independent datasets
- Integrate with gene expression data for multi-omics insights

8. Conclusion

This study successfully identified genome-wide DNA methylation differences between SVD patients and controls using Illumina EPIC array data.

The results demonstrate:

- Significant epigenetic alterations
- Clear group-specific methylation patterns
- Potential for biomarker discovery

Overall, the workflow provides a robust framework for DNA methylation analysis and highlights the importance of epigenetics in disease mechanisms.

9. Files Generated

- DMP_full_results.csv
- DMP_significant.csv
- Hyper_methylated.csv
- Hypo_methylated.csv

- Top_100_CpGs.csv
- Heatmap
- PCA plot
- Volcano plot
- Boxplot

10. References

1. GEO Dataset: GSE318387
2. Aryee et al., *minfi: a flexible and comprehensive Bioconductor package*
3. Ritchie et al., *limma: linear models for microarray data*
4. R Core Team (R Statistical Software)

Submitted by

ANU GOPAL V

MOHAMMAD YAZIN N

YADHU KRISHNAN VB